

Effect of Cerium Reinforcement and Hot-Rolling on Microstructure and Mechanical Properties of High Manganese Steel.

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1. INTRODUCTION

1.1. Background

- Cryogenic applications such as LNG storage tanks demand materials with high strength, ductility, and stability at very low temperatures ($-160\text{ }^{\circ}\text{C}$).
- Conventional materials like austenitic stainless steels and 9% Ni steels face challenges of high cost, magnetization, and weldability issues.



Fig. 1. LNG Storage Tanks

1.1. Background

- High Manganese Steel (HMS) with $>10\%$ Mn has emerged as a promising economical alternative, offering excellent strength, ductility, and cryogenic stability.
- However, long-term service leads to localized corrosion (pitting, intergranular), reducing durability.

1.2. Solution Approach

- Microalloying High Manganese Steel (HMS) with rare earth element Cerium (Ce) and Hot Rolling

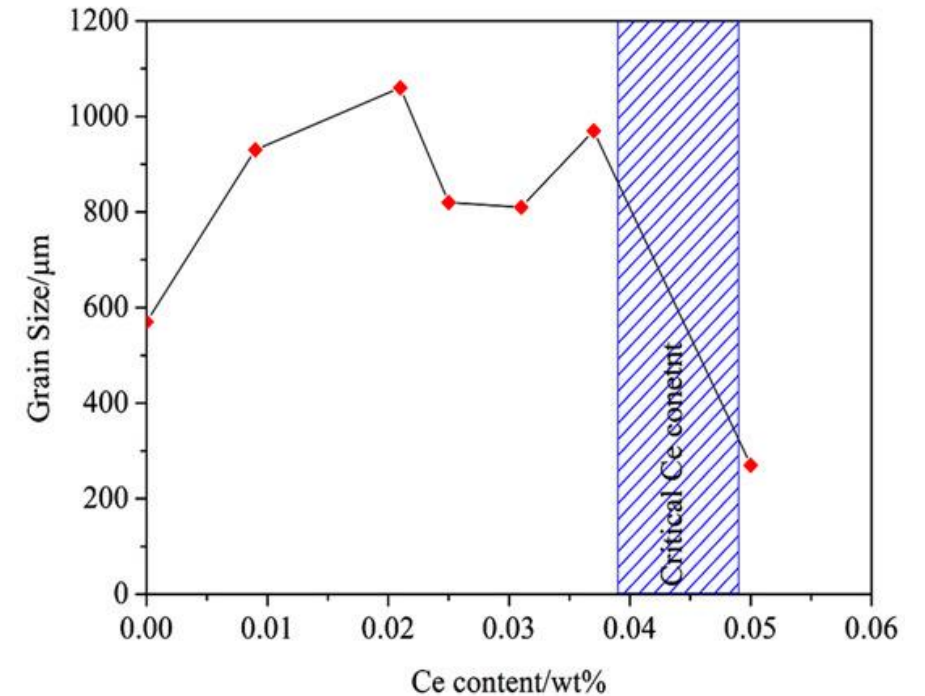


Fig. 2. Relationship between the grain size and cerium content in a cast high manganese austenitic steel [6]

1.3. Why Cerium?

- It refines Prior-Austenite Grains[3].
- It promotes Discontinuous Distribution of Precipitates[3].
- It enhances Spheroidization of Inclusions.[4]
- It increases microstructural stability.[3]

1.4. Why hot rolling?

- Hot rolling further enhances properties through inclusion redistribution, dynamic recrystallization, and microstructural refinement.[5]

2. Literature Review

Alloying Element/Treatment	Steel Type Investigated	Properties Systematically Studied	Key Mechanistic Findings	Ref.
Cerium (Ce)	High Manganese Steel (HMS)	Microstructure, Segregation, Corrosion Resistance	Cerium addition refined grain boundary carbides into fine precipitates and minimized the segregation of Mn, Cr, and C by hindering diffusion due to its large atomic radius. It also promoted the formation of a dense, continuous oxide film that increases resistance, shifting the mechanism from localized pitting to uniform corrosion.	1
Cerium (Ce)	Low Carbon Manganese Steel	Microstructure, Tensile, Yield strength, Elongation, and Charpy Impact Toughness.	Cerium transformed irregular inclusions into fine, spherical Ce-compounds that, due to a low lattice misfit (<6%), acted as effective nucleation sites for grain refinement. This structural enhancement doubles impact energy (to 167 J) by shifting the failure mechanism from brittle cleavage to ductile dimple fracture.	2

2. Literature Review

Alloying Element/Treatment	Steel Type Investigated	Properties Systematically Studied	Key Mechanistic Findings	Ref.
Cerium (Ce)	Cryogenic High Manganese Steel (HMS)	Microstructure, Segregation	Significant refinement occurred above 0.018 wt% Ce, where constitutional supercooling and effective CeAlO ₃ nucleation sites minimize dendrite spacing. This restricted dendrite growth and suppresses the co-segregation of Mn, C, and Si, yielding the finest structure at higher concentrations.	3
Carbon (C) & Manganese (Mn)	High Manganese Austenitic Steel	Microstructure, Yield strength, Hardness, Impact Toughness and Wear Resistance	High carbon favors hardness for low-stress wear, while high manganese is critical for high-stress impact resistance.	4

2. Literature Review

Alloying Element/Treatment	Steel Type Investigated	Properties Systematically Studied	Key Mechanistic Findings	Ref.
Mechanical & Thermal Treatment	SS304L Austenitic Steel	Microstructure, Correlation between twin density, grain size, and hardness.	Significant refinement occurred above 0.018 wt% Ce, where constitutional supercooling and effective CeAlO ₃ nucleation sites minimize dendrite spacing. This restricted dendrite growth and suppresses the co-segregation of Mn, C, and Si, yielding the finest structure at higher concentrations.	5

3. RESEARCH GAP

The study of the effect of Cerium (Ce) reinforcement and hot rolling on the microstructure, mechanical properties and corrosion resistance of high manganese steel has not been conducted systematically.

4. OBJECTIVE

To Investigate the Effect of Cerium Reinforcement and Hot-Rolling on Microstructure and Mechanical Properties of High Manganese Steel

- To improve the mechanical properties, corrosion resistance and achieve refined grains uniformly throughout the microstructure of high manganese steel after reinforcement of Cerium (Ce).

5. METHODOLOGY & EXPERIMENTAL PROCEDURE

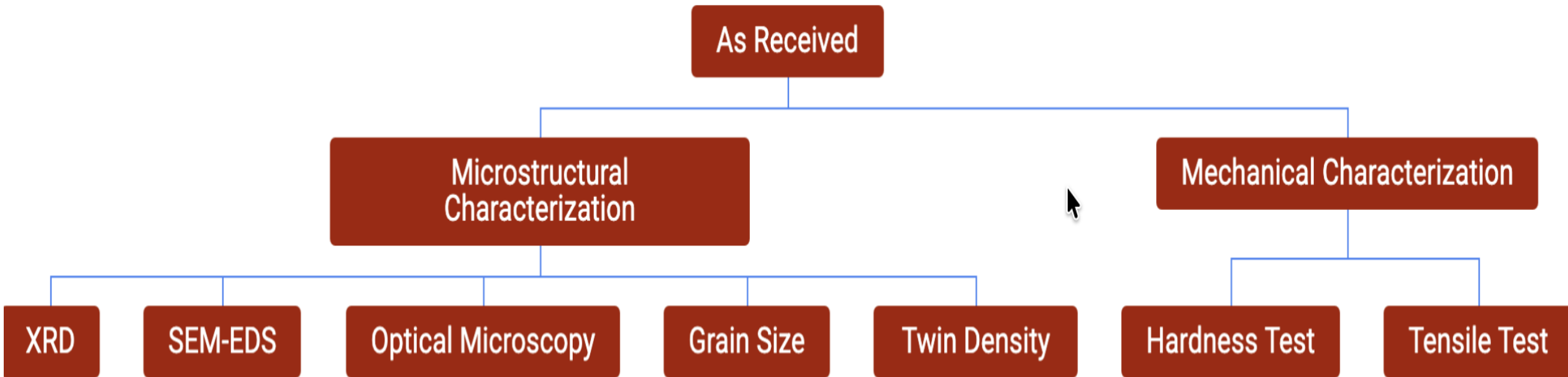


Fig. 3. Schematic representation of the experimental procedure.

5. METHODOLOGY & EXPERIMENTAL PROCEDURE

Table 1. Experimental composition of as-received high manganese steels with varying cerium (Ce) content [8]

Specimen	C	Mn	Cr	Mo	Ni	Si	P	Ce	Fe
Ce-0	0.707	12.5	0.76	0.44	0.304	0.76	0.023	0	Bal.
Ce-0.04	0.703	12.5	0.75	0.42	0.305	0.74	0.024	0.04	Bal.
Ce-0.06	0.705	12.3	0.73	0.46	0.307	0.78	0.025	0.06	Bal.

5.1. Microstructural Characterization

- The as received material was sectioned into specimens of suitable dimensions.
- The specimens were mechanically ground using emery paper ranging from 120 to 2500 grit, followed by cloth polishing with an alumina suspension and a final water polish.
- The polished specimens were subjected to X-ray diffraction (XRD) analysis to determine the phase composition.
- The polished specimens were etched using 5% Nital solution and Optical Microscopy was employed to observe the general microstructure, measure grain sizes and twin density.
- The etched specimens were examined using Scanning Electron Microscopy (SEM) to investigate the detailed morphology and the distribution of inclusions.

5.2. Mechanical Characterization

- The polished and etched specimens, prepared earlier, were subjected to Vickers hardness testing to evaluate the mechanical hardness.
- Specimens for tensile testing are currently being prepared at Cental Institute of Petrochemicals Engineering & Technology (CIPET), Jaipur, to determine the material's tensile properties.

6. RESULTS & DISCUSSION

6.1. Phase Analysis using X-ray Diffraction

Figure 4. shows the XRD patterns of the polished specimen. Five major peaks had been indexed and identified as peaks corresponding to (111), (200), (220), (311) and (222) planes of face-centred cubic (fcc) phase. The diffraction pattern clearly reveals the presence of only single-phase face-centred cubic (fcc) crystal structure, i.e. austenite.

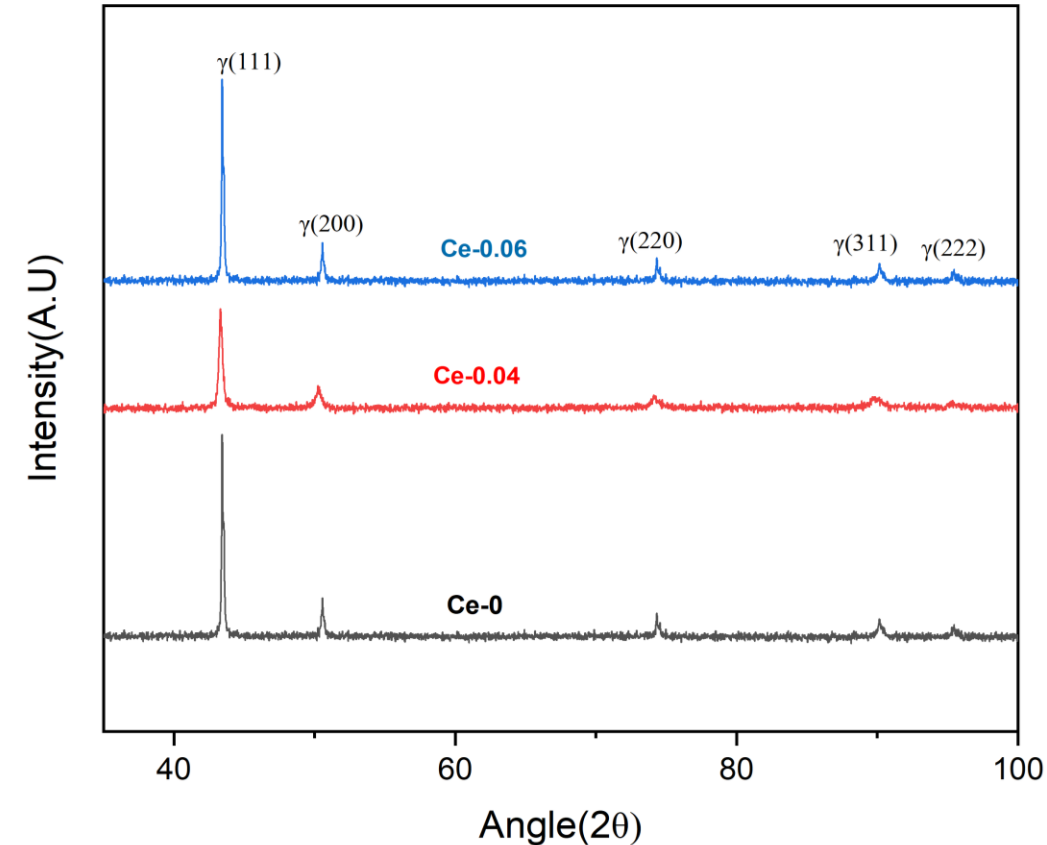


Fig. 4. X-ray diffraction pattern of polished specimen.

6.2. Microstructural Analysis

- The optical micrographs of the specimens with varying Cerium (Ce) contents are presented in Figure 5. The microstructure of all the specimen exhibits a fully **austenitic matrix** composed of **equiaxed polyhedral grains**.
- Distinct deformation twins were observed within the austenite grains across all compositions, indicative of the low stacking fault energy (SFE) inhibiting dislocation cross-slip, forcing the material to accommodate the high compressive strains of hot rolling through mechanical twinning.

6.2. Microstructural Analysis

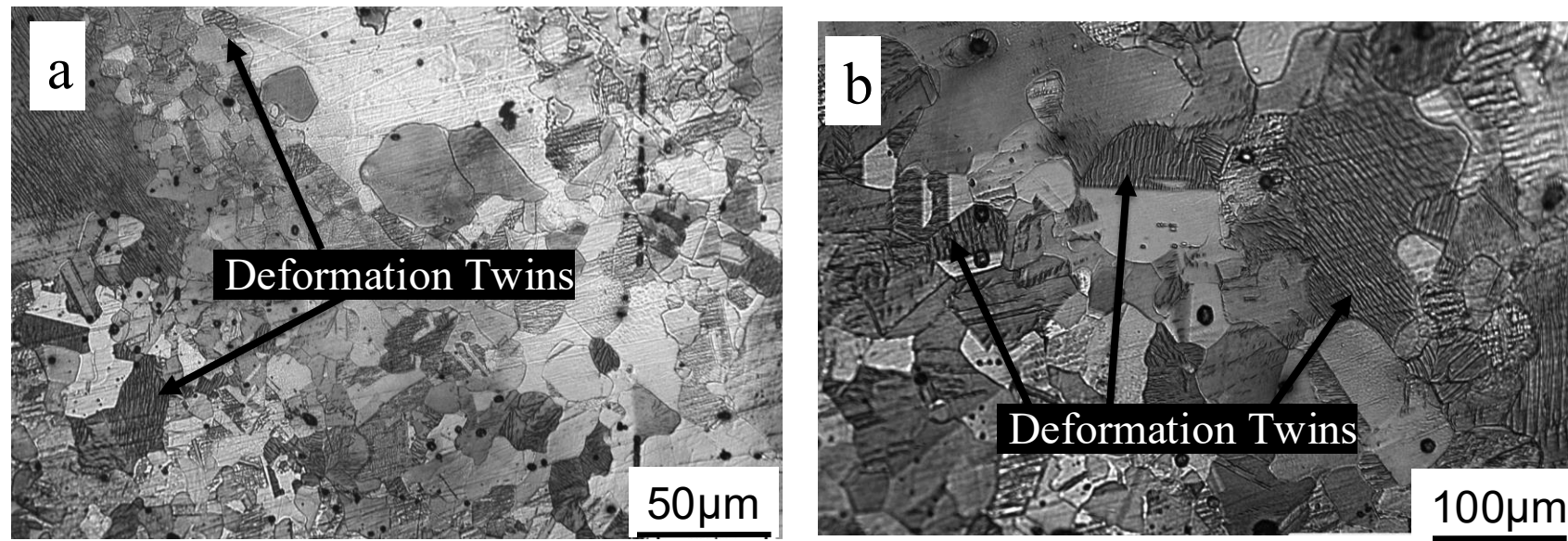


Fig. 5. Optical Micrographs of Ce-0 at (a) low magnification and (b) high magnification.

6.2. Microstructural Analysis

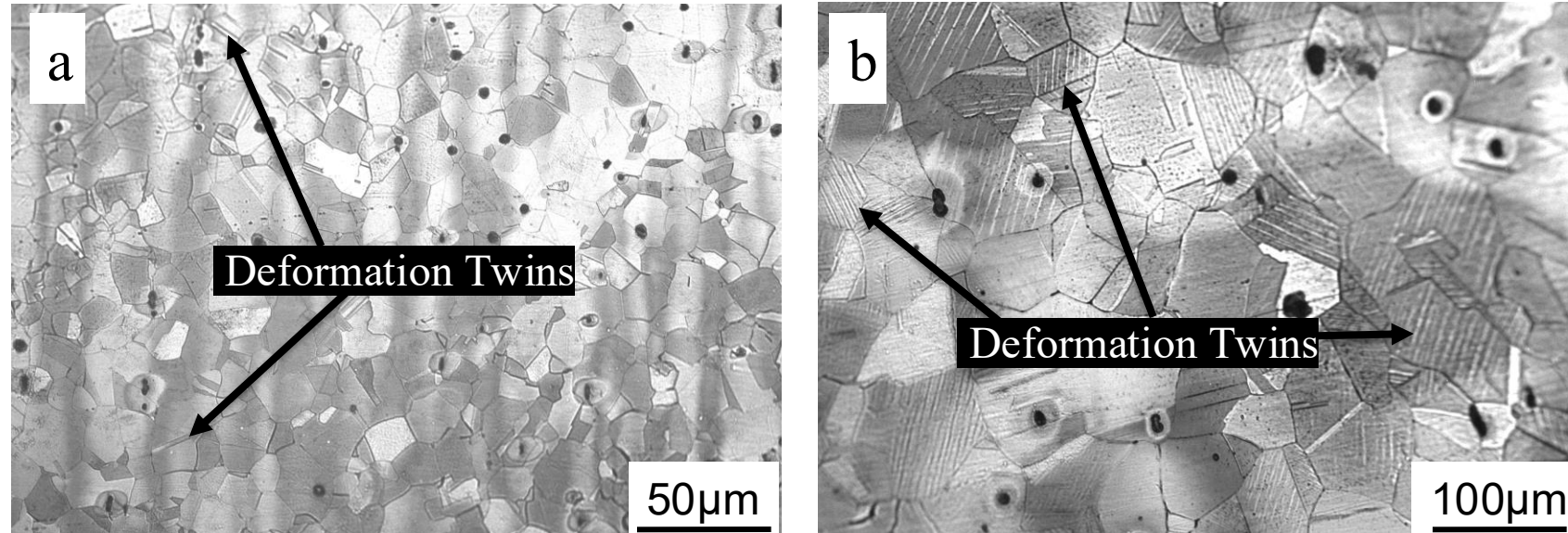


Fig. 6. Optical Micrographs of Ce-0.04 at (a) low magnification and (b) high magnification.

6.2. Microstructural Analysis

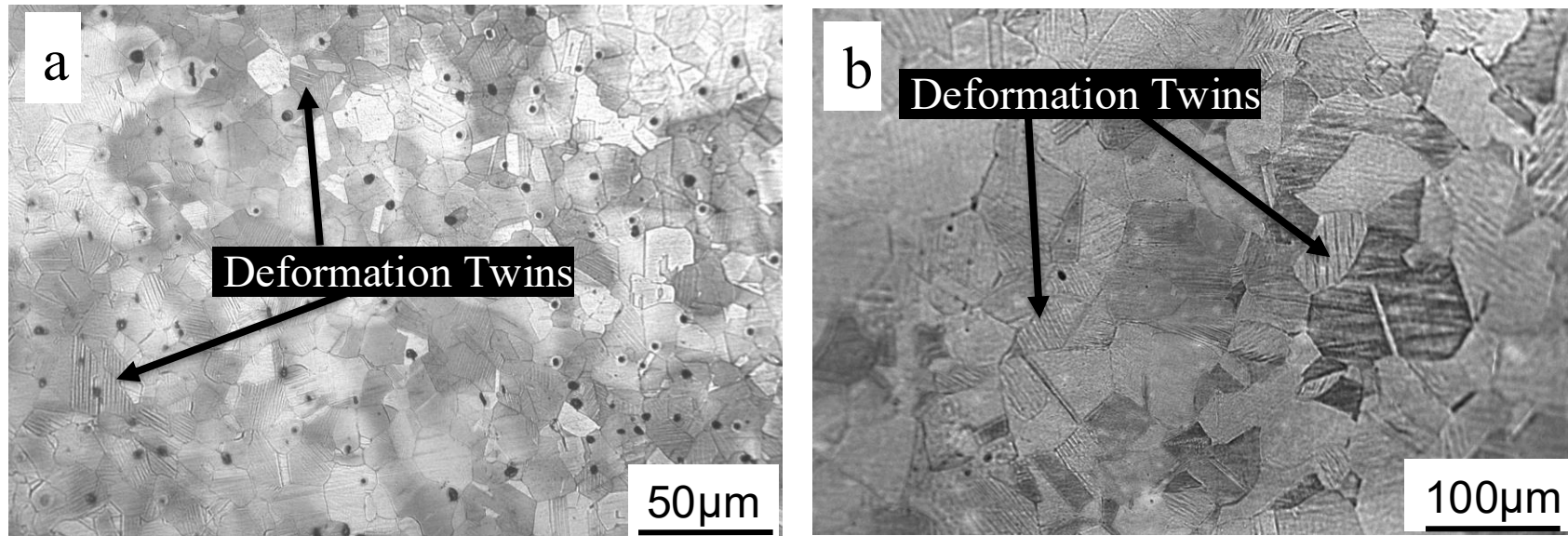


Fig. 7. Optical Micrographs of Ce-0.06 at (a) low magnification and (b) high magnification.

6.3. SEM – EDS Analysis

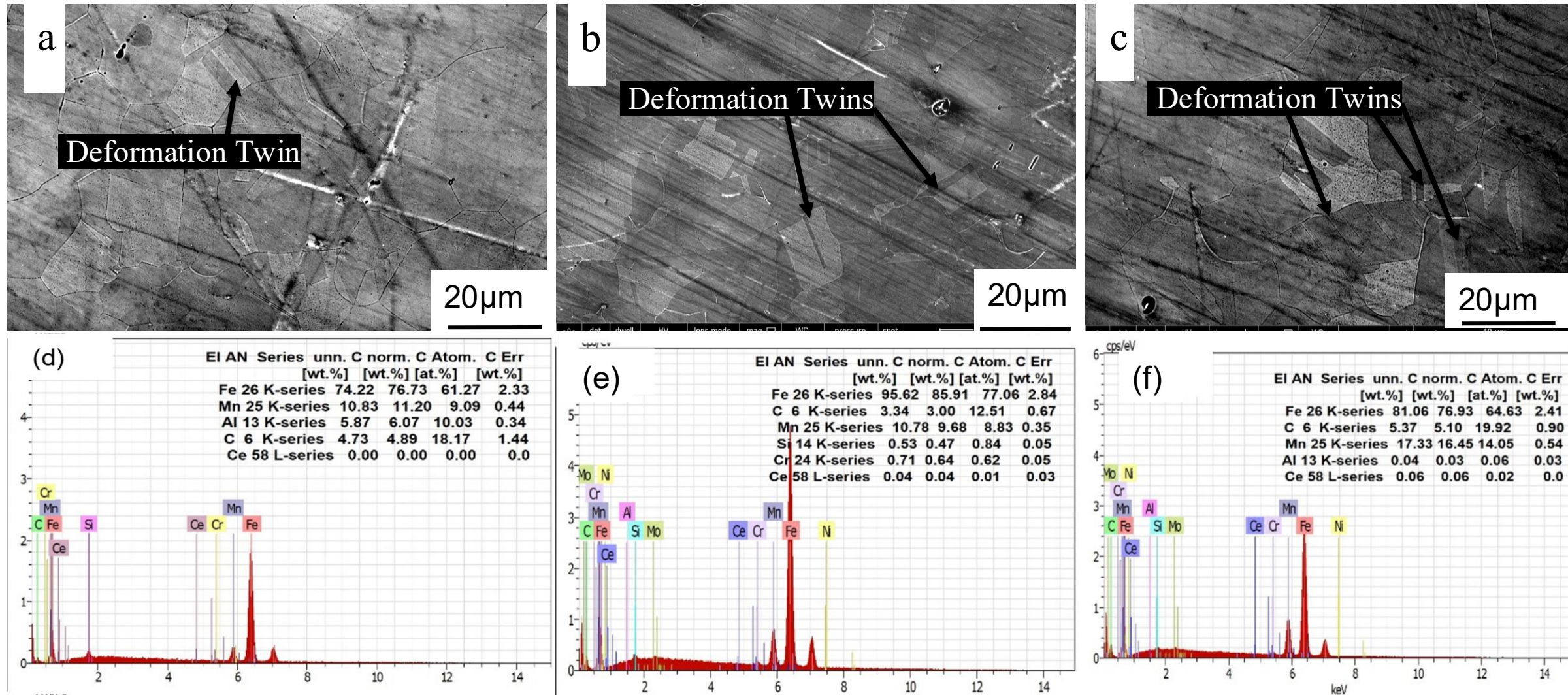


Fig. 8. SEM images of (a) Ce-0, (b) Ce-0.04, (c) Ce-0.06 and elemental mapping analysis of (d) Ce-0, (e) Ce-0.04, (f) Ce-0.06 specimens

6.3. SEM – EDS Analysis

- The SEM micrographs reveal a distinct refinement in the austenitic microstructure, where the average grain size progressively decreased with increasing Cerium content due to the element's effective grain boundary pinning, heterogeneous nucleation effects as well as hot rolling. Concurrently, the twin density within these finer grains exhibited a marked increase.
- Energy Dispersive Spectroscopy (EDS) analysis confirmed the successful mixing and dispersion of Cerium throughout the refined microstructure.

6.4. Hardness and Grain Size Correlation

- Figure 7. illustrates the simultaneous variation of Vickers microhardness and average grain size for the high manganese steel specimens with varying Cerium content. A distinct **inverse correlation** is observed between the two parameters.
- The addition of Cerium appears to promote grain boundary pinning, leading to a more uniform and slightly finer grain size distribution compared to the base alloy.

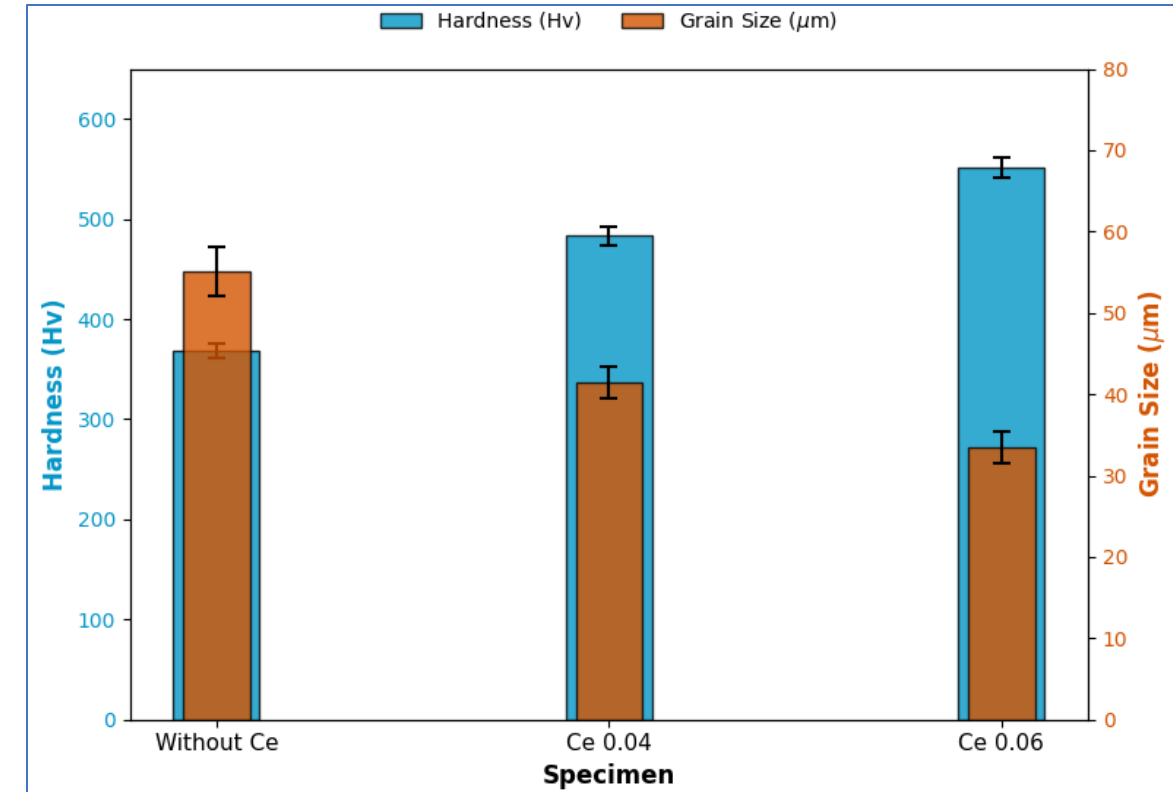


Fig. 9. Average hardness value and average grain size of Ce-0, Ce-0.04 and Ce-0.06 specimens.

6.5. Hardness, Twin Density and Grain Size Correlation

Table 2. Summary of average grain size, twin density, and Vickers microhardness for the investigated high manganese steel specimens.

S.No.	Specimen	Avg. Grain Size (μm)	Avg. Hardness (Hv)	Avg. Twin Density (mm ⁻¹)
1	Ce-0	55.05 ± 3	368 ± 7	145.71
2	Ce-0.04	41.45 ± 2	483 ± 9	168.73
3	Ce-0.06	33.43 ± 2	552 ± 10	187.36

6.5. Hardness, Twin Density and Grain Size Correlation

- The quantitative data presented in Table 2. indicates a distinct inverse relationship between the average grain size and the mechanical microhardness. With the addition of Cerium (0 to 0.06 wt.%), the average grain size decreased significantly from $55.05 \pm 3 \mu\text{m}$ to $33.43 \pm 2 \mu\text{m}$. Concurrently, the twin density exhibited a monotonic increase from 145.71 mm^{-1} to 187.36 mm^{-1} .
- The substantial increase in hardness from $368 \pm 7 \text{ Hv}$ (Ce-0) to $552 \pm 10 \text{ Hv}$ (Ce-0.06) is primarily governed by grain boundary strengthening, adhering to the classical **Hall-Petch relationship**:

$$(\sigma_y = \sigma_0 + k \cdot d^{-1/2})$$

where σ_y is the yield stress, σ_0 is the friction stress, d is grain diameter and k is strengthening coefficient.

6.5. Hardness, Twin Density and Grain Size Correlation

- The refinement of grains due to Cerium and Hot Rolling introduces a higher density of grain boundaries, which act as effective obstacles to dislocation motion, thereby increasing the stress required for plastic deformation.
- Furthermore, the increased twin density contributes significantly to the hardening mechanism. Analogous to grain boundaries, twin boundaries serve as effective barriers to dislocation slip (often referred to as the "Dynamic Hall-Petch effect").
- The reduction in the mean free path of dislocations, caused by the synergistic effect of grain refinement and increased twin density, results in the superior hardness observed in the Ce-modified specimens.

7. Conclusions

The following conclusions can be drawn from the present work:

- The X-ray diffraction (XRD) analysis revealed that all the developed high manganese steel specimens exhibited a stable **single-phase face-centered cubic (FCC) austenitic structure**. It was observed that the addition of Cerium (up to 0.06 wt.%) did not induce the formation of any deleterious secondary phases, thereby preserving the phase stability essential for cryogenic applications.
- The microstructural examination demonstrated that the reinforcement of Cerium resulted in a significant refinement of the prior-austenite grains. This refinement was attributed to the solute drag effect and the pinning of grain boundaries by Cerium-based inclusions, which effectively restricted grain growth.

7. Conclusions

The following conclusions can be drawn from the present work:

- The specimens exhibited a distinct presence of twins, wherein the twin density increased with increasing Cerium content. The extensive formation of these twins was attributed to the low Stacking Fault Energy (SFE) of the high manganese steel matrix, which energetically favors the formation of twin boundaries during the thermal processing.
- The average Vickers microhardness values increased significantly with Cerium addition. This enhancement in hardness was correlated with the Hall-Petch relationship, where the refined grain structure and the increased density of twin boundaries (Dynamic Hall-Petch effect) acted as effective barriers to dislocation motion, thereby requiring higher stress for deformation.

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THANK YOU

